

**KARNATAK UNIVERSITY  
DHARWAD**



**Janata Shikshana Samiti's  
JSS Shri Manjunatheshwara Institute of UG & PG  
Studies, Vidyagiri, Dharwad-580 004.**



**A PROJECT REPORT ON  
“Cyber Café Management System”  
BACHELOR OF COMPUTER APPLICATIONS (BCA)  
OF  
KARNATAK UNIVERSITY, DHARWAD**

**PROJECT GUIDED BY**

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**Submitted By**

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**BCA VI SEMESTER**

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**DEPARTMENT OF COMPUTER SCIENCE 2024-2025**

**JANATA SHIKSHANA SAMITI'S**  
**JSS SHRI MANJUNATHESHWARA INSTITUTE OF UG**  
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**CERTIFICATE**

*This is to certify that **Ms. Dhanya Hegde** has satisfactorily completed Project Work entitled “**Cyber Café Management System**” for the partial fulfilment of BCA prescribed by **Karnatak University Dharwad** during the academic year **2024-2025**.*

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Examiners:

1)

2)

## **ACKNOWLEDGEMENT**

*The successful presentation of this project is acknowledgements of the immense support extended by JSS SMI UG & PG STUDIES, DHARWAD, which has provided us opportunity to fulfil the most cherished & desired way to reach my goal.*

*I would like to express my heartfelt thanks to our President, Shri Shri Vishwaprasanna Theertha Swamiji of Sri Pejavarmath, Udupi, Parama Pujya Dr. D. Veerendra Heggade ji, the Chairman of Janata Shikshana Samiti & Dharmadhikari of Dharmasthala.*

*I would like to express my sincere and hearty thanks of gratitude to our beloved Principal Dr. Ajith Prasad for giving constant support and valuable time.*

*I would also like to express my gratitude to our Head of the Computer Science Department Prof. Vivek M. Laxmeshwar who gave me the golden opportunity to do this wonderful project on the topic “CYBER CAFÉ MANAGEMENT SYSTEM”, which also helped me in doing a lot of research and exposed to lot of new information that would help me in my mere future.*

*I would also take this opportunity to offer my sincere gratitude to my Project Guide Prof. Narasimha Bhat for his excellent support throughout the development of this project and providing the necessary information on my demand at all the times.*

**Dhanya Hegde**

## **DECLARATION**

*I, Miss. Dhanya Hegde, Student of Sixth Semester BCA,  
Department of Computer Science, JSS SHRI MANJUNATHESHWARA  
INSTITUTE OF UG & PG STUDIES, VIDYAGIRI, DHARWAD affiliated to  
Karnatak University, Dharwad hereby declare that the Project entitled  
“Cyber Café Management System” submitted in partial fulfilment of  
the course requirement for the Award of Degree in Bachelor of  
Computer Application, Karnatak University, Dharwad during the  
Academic Year 2024-2025. I have not submitted the matter embodied  
to any other University or Institution for the Award of any other  
Degree.*

Date:

Dhanya Hegde

Place: Dharwad

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# **1.SYNOPSIS**

## **1.1 Title of the project**

### **Cyber Café Management System**

Cybercafé management system is a software package, which is intended to be used in cyber cafes for managing the clients' computer efficiently. Now a day's cyber terrorism, which is mainly undergone through internet cafes, need to be tackled properly. Thereby, it is indeed necessary to store the valid information of the user who comes for internet access. The system being used, the time at which the user logs in and logs out should be recorded systematically.

## **1.2 Objectives of the project**

- The project aims to automate the management processes of a Cybercafé, including User registration, login tracking, time usage monitoring, billing & generating reports.
- Aiming to improve efficiency, reduce errors, & provide a user-friendly interface for both User and admin.
- Accurately monitor customer internet usage time on individual computers. Calculate and generate bills based on time spent online.
- Number of people access the internet frequently by means of cyber cafes. For such frequent users, discounted rates may be charged from them.

## **1.3 Input of the project**

- User information:
  - User name, contact number, email address, id proof details (if required), fees.
  - Login and logout time

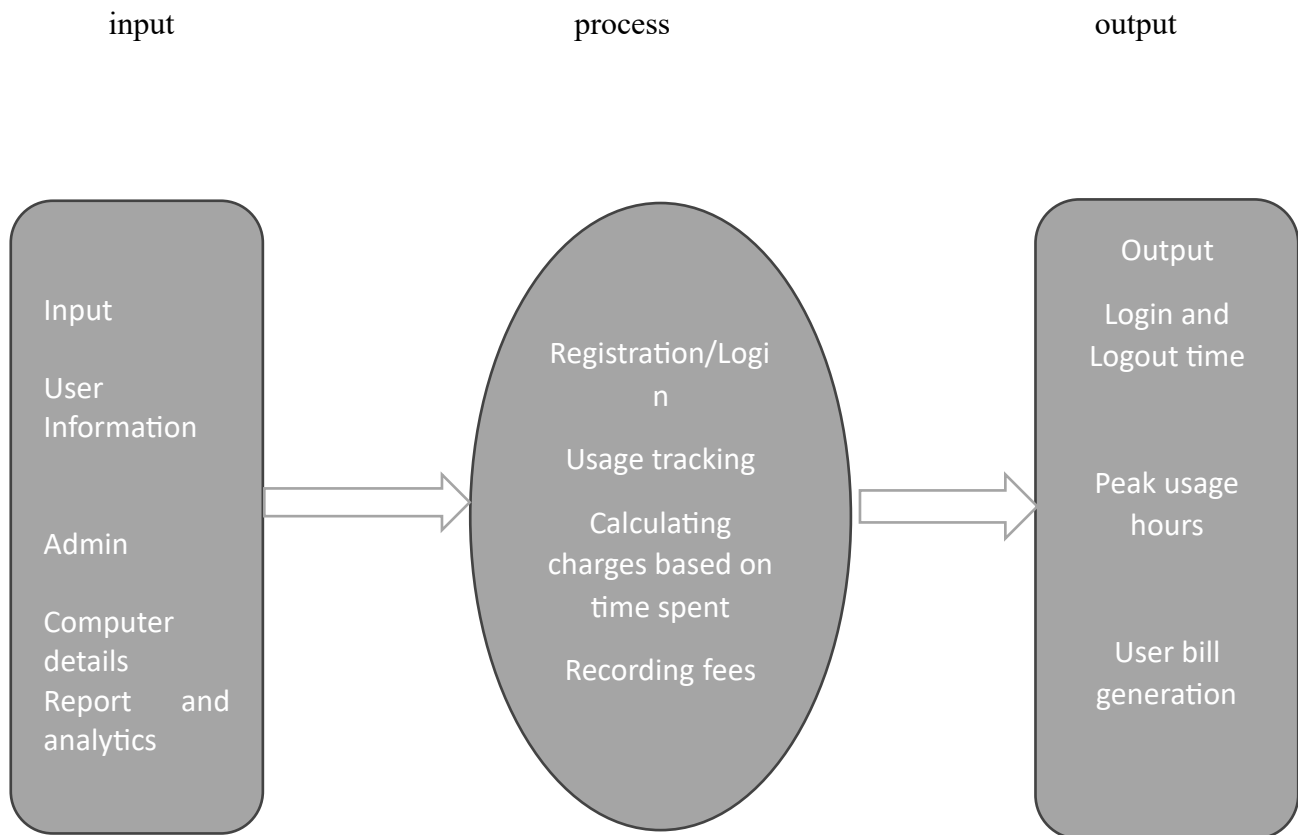
- Admin information:
  - Admin name, id, password
  - Username, mobile number, email address
  - Computer station number
- Computer details:
  - Computer name, IP address
  - Computer Availability

### **1.4 Output of the project**

- User bill:
  - User details, Total amount, Date/time
- Usage report:
  - Daily/Weekly/monthly  
Revenue
  - Computer usage  
statistics
- Usage logs:
  - User detail
  - Login time
  - Logout time



### **1.5 Process logic**



### **1.6 Existing system**

- ✧ It refers to a manual method of managing user access, billing, and usage, often involving paper-based records and manual time tracking.

### **1.7 proposed system**

- ✧ This would be a computerized management system that automates these processes, providing features like user logins, real time usage monitoring, automated billing, significantly improving efficiency and accuracy compared to manual methods.

### **1.8 Tools/ Platform, Language to be used**

#### **○ Hardware Requirements:**

1. Hard disk :500GB
2. Processor: Core i5
3. RAM: 8GB

#### **○ Software Requirements:**

1. Operating Systems: Windows, Linux
2. Frontend: HTML, CSS, JavaScript
3. Backend: MySQL, PHP
4. Server: WAMP, XAMPP

### **1.9 Are you doing project for any industry/ client:**

No

### **1.10 Duration of the project:**

- 2 Months

### **1.11 Limitation of the project:**

- Potential for security breaches if not properly implemented.
- Limited functionality for complex customer needs, dependence on reliable internet connection.
- If the system crashes or the internet goes down, operation can be distributed.

### **1.12 Scope of the project:**

- As this is automated cyber management system it makes it very easy for administrator to search details.
- Time allotment is done very efficiently.
- It provides fast service in term of the calculation and time management.
- People travelling to areas without reliable internet access will use cyber cafes.
- We can add online payment method.

## **2.INTRODUCTION**

The Cyber Cafe Management System is a web-based software project developed to simplify and manage the routine operations of a cyber café. It aims to digitize the manual process of handling user registrations, computer allocations, and session tracking, thereby improving efficiency and user experience.

This system includes key modules such as Admin, User, and Computer. The Admin module is responsible for managing user accounts, monitoring active sessions, and maintaining system records. The User module allows customers to register, log in, and manage their profiles. The Computer module helps in tracking system availability and managing hardware details such as monitor, keyboard, and mouse.

By centralizing all user and computer data, the system reduces administrative effort, eliminates the need for physical record-keeping, and minimizes errors. It also enhances transparency and allows admins to make informed decisions based on real-time data.

The current version focuses on the core functionalities required for effective cyber café management. Modules like Payment and Invoice are planned for integration in the future to enhance the system's capabilities and provide end-to-end digital management. These additions will support features such as automated billing, session-based charging, and invoice generation, making the system more complete and business-ready.

Overall, this system aims to bring convenience, accuracy, and professionalism to the management of cyber café services.

## **Introduction to technologies used in this project:**

The Cyber Cafe Management System is developed using a combination of front-end and backend technologies that work together to deliver a dynamic and user-friendly experience. The front end is built with HTML and CSS, which are used to design and structure responsive and visually appealing web pages. JavaScript is used to add interactivity and enhance the user experience, such as form validation and dynamic content updates.

For the backend, PHP is used as the server-side scripting language to handle business logic, form processing, and database interactions. MySQL serves as the database management system, where all critical data such as user details, computer information, and login credentials are securely stored. The application is developed and tested using the WAMP Server, which provides an integrated environment consisting of Windows, Apache, MySQL, and PHP, making local development easier and more efficient.

This technology stack ensures that the system is lightweight, easy to maintain, and capable of handling the essential operations required in a cyber café setup.

### **HTML:**

HTML means Hypertext Markup Language. HTML is a method of describing the format of document, which allows them to be viewed on computer screen. Web browsers display HTML documents, program which can navigate across networks and display a wide variety of types of information. HTML pages can be developed to be simple text or to be complex multimedia extra advantages containing, moving images, virtual reality, and java applets. The global publishing format of the Internet is HTML. It allows authors to use not only text but also format that text with headings, list and tables, and also includes still images videos, and sound within text. Readers can access pages information from any where in the world at the click of mouse button information can be downloaded to readers own PC or workstations HTML pages can also be used for entering a data and as a front end for commercial transaction.

Example to display message using HTML tags

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
```

```
"http://www.w3.org/TR/html4/loose.dtd">
```

```
<html>
```

```
<head>
```

```
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">
```

```
<title>Demo html</title>
```

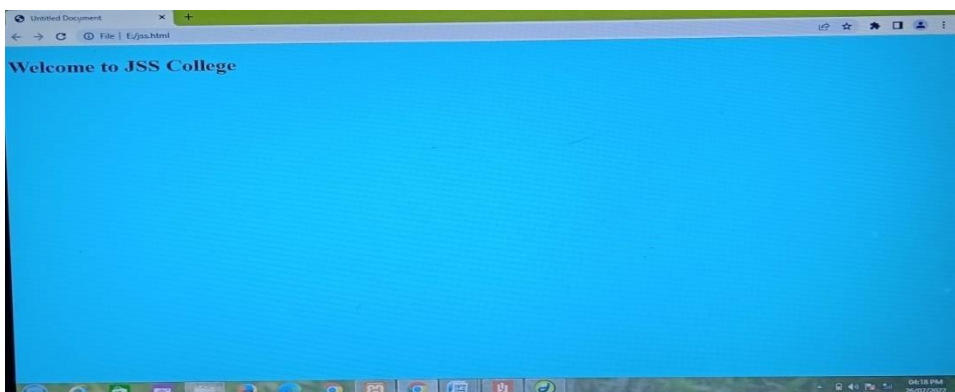
```
</head>
```

```
<body>
```

```
<h1>Welcome to JSS College</h1>
```

```
</body>
```

```
</html> Output:
```



## **Why you need WAMP, MYSQL and PHP?**

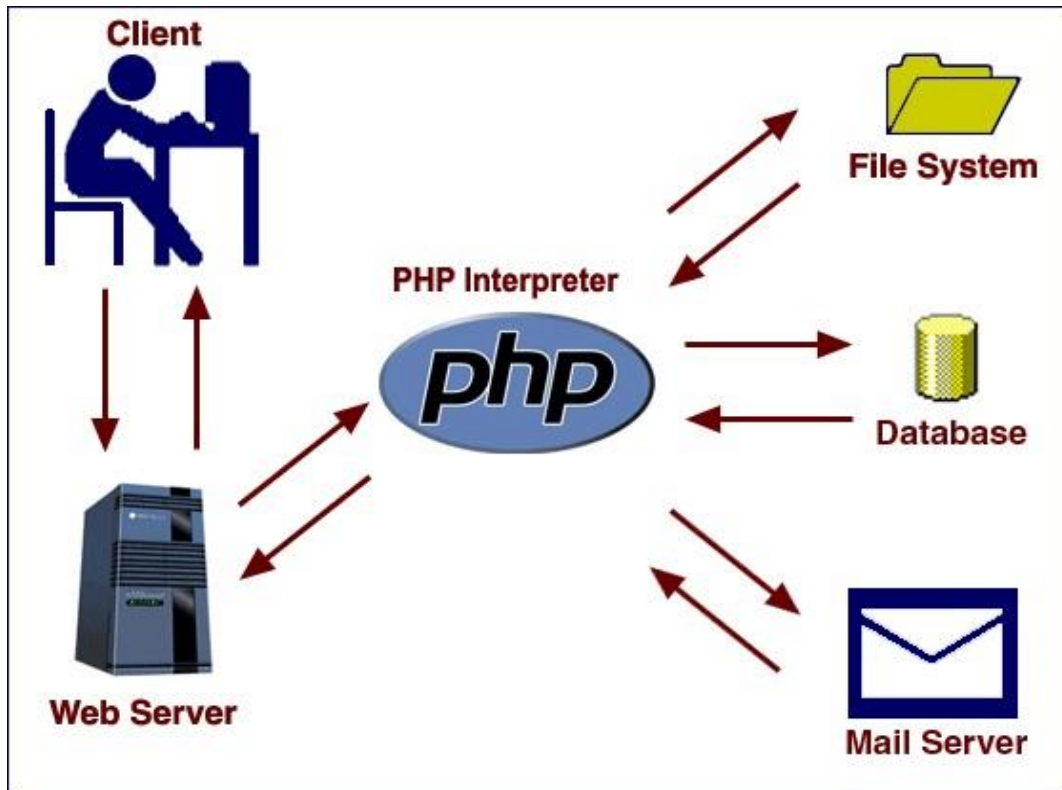
PHP is a powerful scripting language that can be run by itself in the command line of any computer with PHP installed. However, PHP alone is not enough in order to build dynamic web sites. To use PHP on a web site, you need a server that can process PHP scripts. WAMP server allows developers to test PHP scripts locally. Additionally dynamic websites are dependent on stored information that can be accessed easily; this is the main difference between a dynamic site and a static HTML site. However, PHP does not provide a simple, efficient way to store data. This is where a relational database management system like MySQL comes into play.

### **PHP:**

PHP is a scripting language originally designed for producing dynamic webpages. It has evolved to include a command line interface capability and can be used in standalone graphical application. While PHP was originally created by Rasmus Lerdorf in 1995, the main implementation of PHP is now produced by the PHP Groups and serves as the de facto standard for PHP as there is no formal specification.

PHP is a scripting language under the PHP License; however, it is incompatible with the GNU General Public License (GPL). Due to restrictions on the usage of the term PHP. It is widely used general-purpose scripting language that is especially suited for web development and can be embedded into HTML. It generally runs on a web server, taking PHP code as its input, I am creating web pages as out puts. It can be deployed on web servers and on almost every operating system and platform free of charge. PHP in installed on more the twenty million web sites and one million web servers.

## PHP Architecture:



## Usage:

PHP primarily acts as a filter, taking input from a file or stream containing text and/or PHP instructions and outputs another stream of data; most commonly the output will be HTML. It can automatically detect the language of the user. From PHP 4, the PHP parser compiles input to produce byte code for processing by the Zend Engine, giving improved performance over its interpreter predecessor. Originally designed to create dynamic web pages, PHP's principal focus is server-side scripting, and it is similar to other server-side scripting languages that provide dynamic content from a web server to a client, such as Microsoft's Active Server Pages, Sun Microsystems Java Server Pages, and mod\_perl. PHP has also attracted the development of many frameworks that provide building blocks and a design structure to promote rapid application development (RAD). Some of these include CakePHP, Symfony, CodeIgniter, and Zend Framework, offering features similar to other web application frameworks.



The WAMP architecture has become popular in the web industry as a way of deploying web applications. PHP is commonly used as the PHP in this bundle alongside Linux, Apache and MySQL, although they may also refer to Python or Perl. As of April 2007, over 20 million Internet domains were hosted on servers with PHP installed, and PHP was recorded as the most popular Apache module. Significant websites are written in PHP including the user-facing portion of Facebook, Wikipedia (Media Wiki), Yahoo!, My Yearbook, WordPress. In addition to server-side scripting, PHP can be used to create stand-alone, compiled applications and libraries, it can be used for shell scripting.

## MY SQL:

### How to pronounce “MySQL”:

“My ess-cue-el” is the “official” way to pronounce “MySQL,” but pronouncing it “my sequel” is common too.

## WHAT IS MYSQL DATABASE?

**MYSQL** is a freely available open-source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL). SQL is the most popular language for adding, accessing and managing content in a database. It is most noted for its quick processing, proven reliability, ease and flexibility of use.

### FEATURES OF MYSQL:

**1.Relational Database System:** Like almost all other database systems on the market, MySQL is a relational database system.

**2.Client/Server Architecture:** MySQL is a [client/server system](#). There is a database server (MySQL) and arbitrarily many clients (application programs), which communicate with the server; that is, they query data, save changes, etc. The clients can run on the same computer as the server or on another computer (communication via a local network or the Internet).

## ADVANTAGES OF MYSQL:

**1.It is easy to use:** While a basic knowledge of SQL is required—and most relational databases require the same knowledge—MySQL is very easy to use. With only a few simple SQL statements, you can build and interact with MySQL.

**2.It is secure:** MySQL includes solid data security layers that protect sensitive data from intruders. Rights can be set to allow some or all privileges to individuals. Passwords are encrypted.

**3.It is inexpensive:** MySQL is included for free with NetWare® 6.5 and available by free download from [MySQL Web site](#).

**4. It is fast:** In the interest of speed, MySQL designers made the decision to offer fewer features than other major database competitors, such as Sybase and Oracle. However, despite having fewer features than the other commercial database products, MySQL still offers all of the features required by most database developers.

**5. It's scalable:** MySQL can handle almost any amount of data, up to as much as 50 million rows or more. The default file size limit is about 4 GB. However, you can increase this number to a theoretical limit of 8 TB of data.

**6.It runs on many operating systems:** MySQL runs on many operating systems, including Novell NetWare, Windows\* Linux\*, many varieties of UNIX\* (such as Sun\* Solaris\*, AIX, and DEC\* UNIX), OS/2, FreeBSD\*, and others.

**7.It manages memory very well:** MySQL server has been thoroughly tested to prevent memory leaks.

## Security:

Views are basically used as a part of security, means in many organizations, the end user will never be given original tables & all data entry will be done with the help of views only. But the database administrator will be able to see everything because all the operations done by the different users will come to the same table.

## Queries:

A query is a question or a request. With MySQL, we can query a database for specific information and have a record set returned. Create a connection to a database: Before you can access data in a database, you must create a connection to the database. In PHP, this is done with the MySQL connect () function.

Syntax:

MySQL connect (server name, username, password);

Server name: Optional Specifies the Server to connect. Default values is localhost: 3306

## MySQL Functions:

What is a database? Quite simply, it's an organized collection of data. A database management system (DBMS) such as Access, FileMaker Pro, Oracle or SQL Server provides you with the software tools you need to organize that data in a flexible manner. It includes facilities to add, modify or delete data from the database, ask questions (or queries) about the data stored in the database and produce reports summarizing selected contents. MySQL is a multithreaded, multiuser SQL database management system (DBMS). The basic program runs as a server providing multi-user access to a number of databases. Originally financed in a similar fashion to the JBoss model, MySQL was owned and sponsored by a single for-profit firm, the Swedish company MySQL AB now a subsidiary of Sun Microsystems, which holds the copyright to most of the codebase. The project's source code is available under terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL is a database. The data in MySQL is stored in database objects called tables. A table is a collection of related data

entries and it consists of columns and rows. Databases are useful when storing information categorically.

### **Why phpMyAdmin is Ideal for My Project:**

- **Efficient Data Handling:** Easily browse and update complex User, computer and usage records.
- **Visualization of Relationships:** Understand and manage links between tables using foreign key relationships .
- **Time-Saving:** Quick access to insert, update, and fetch data during development without writing full SQL each time.
- **Secure Role-Based Access:** With tables like security and users, phpMyAdmin helps verify and manage role-based database entries.
- **Export/Import Support:** Simplifies backup and transfer of your educational data during upgrades or server migration.
- **Testing & Debugging:** Ideal for testing backend logic, triggers, or queries before integrating with your PHP frontend.

### **3.SYSTEM ANALYSIS**

#### **Existing System**

In the current scenario, many cyber cafés rely on manual methods or basic tools like notebooks, Excel sheets, or simple timers to manage their daily operations. User registrations, login/logout times, and computer allocations are often recorded by hand, which makes the process inefficient and error-prone. Tracking session durations, calculating usage time, and maintaining user history becomes increasingly difficult as the number of users grows. Additionally, the manual system lacks data security, as physical records can be lost, damaged, or misused. There is no proper way to analyse user behaviour, generate reports, or retrieve historical data easily.

Computer availability also needs to be checked physically, which leads to confusion during peak hours and slows down the service. There is no automated mechanism for identifying which systems are in use, which users are logged in, or how long a session has lasted. Moreover, managing multiple users simultaneously becomes a challenge, especially when there is high customer traffic. Communication between staff and users may also become disorganized due to the lack of a proper system interface.

Overall, the existing manual system is time-consuming, difficult to scale, and lacks transparency and reliability. It not only affects the efficiency of the café staff but also impacts customer satisfaction and the overall professionalism of the service.

## Proposed System

The Cyber Cafe Management System aims to overcome the challenges faced by the existing manual system by introducing a centralized, digital solution. The proposed system is a webbased platform that allows admins to efficiently manage users, track computer usage, and monitor system status in real time. It provides a secure login for users and admins, ensuring that data is only accessible to authorized personnel. The system stores all data—such as user details, session logs, and computer hardware info—in a structured MySQL database, making it easy to retrieve and update information when needed.

The admin can view and manage registered users, assign computers, check the availability of systems, and monitor active sessions directly from the dashboard. This digital transformation reduces paperwork, minimizes errors, and saves time. It also enables better decision-making through organized records and potential analytics. The system is designed to be scalable and user-friendly, and future upgrades like payment handling and invoice generation can be easily integrated. Overall, the proposed system improves accuracy, security, transparency, and operational efficiency in managing cyber café services.

## **4.SYSTEM DESIGN**

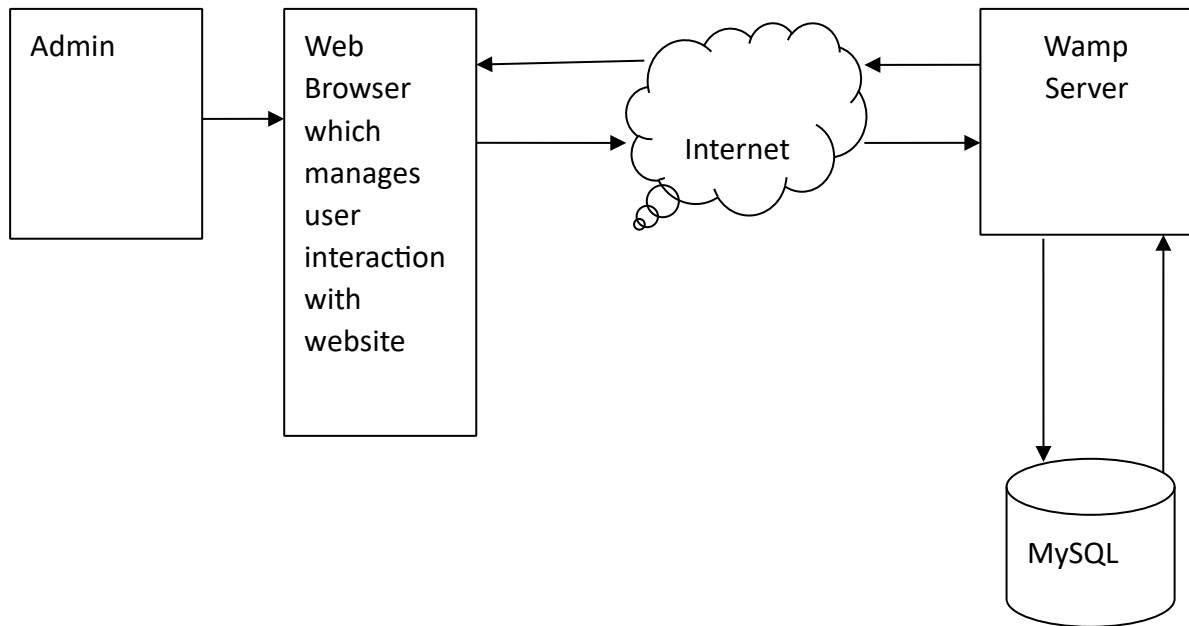
The purpose of the decision phase is to plan a solution of the problem specified by the requirements document. This phase is the first step in moving the problem domain to the solution domain. It involves the process, in which conceiving, planning and carrying out the plan generating the necessary report, In other words, the design phase act as a bridge between SRS and implementation phase. The design of the system is perhaps the most critical factor affecting the quality of the software, and as a major impact on the later phase, particularly the testing and maintenance.

### **Software Design**

Design is the key phase of any project. It is the first step in moving from the problem domain to the solution domain. The input to the design phase is the specifications of the system to be designed. The output of the top-level design is the architectural design, or the system design for the software system to be built. A design should be very clear, verifiable, complete, traceable, efficient and simple.

### **Architecture Design**

The architecture design defines the relationship among major structural element of the program. Architecture diagram shows the relationship between different components of system. This diagram helps to understand the overall concept of system.



## Logical design

The graphical representation of systems' data and how the process transforms the data is known as Data Flow Diagram. It shows the logical flow of the data.

The logical design describes the detailed specification for the system, describing its features, an effective communication and the user interface requirements. The logical system of proposed system should include the following.

1. External system structure.
2. Relationship between all the activities.
3. The physical construction and all the activities.
4. Global data.
5. Control flow.
6. Derived program structure.



## Design Principles

Basic design principles that enable the software engineer to navigate the design process are:

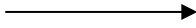
- The design process should not suffer from “Tunnel vision”.
- The design should be traceable to analysis model.
- The design should not reinvent the wheel.
- The design should minimize the intellectual distance between the software and the problem, as it exists in the real world.
- The design should exhibit uniformity and integrity.
- The design should be structured to accommodate changes.
- The design not coding, the coding is not a design.
- The design should be assessed for the quality, as it is being created, not after the fact.
- The design should be reviewed to minimize the conceptual errors.

## DATA FLOW DIAGRAM

The data flow diagram(DFD) is one of the important modelling tools. It shows the user of the data pictorially. DFD represents the flow of the data between different transformations and processes in the systems. The data flow diagram shows logical flow of the data. It represents the functional dependencies within a system. It shows output values in a computation or derived from input values. It is a simple pictorial representation or model for system behaviour. It specifies, “what is to be done but not how is to be done”. It describes the logical structure of the system. It relates data information to various processes of the system. It follows top-down approach.

## Data Flow Diagram Notations

### Data Flow:



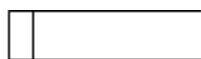
It may be from file-to-file or file-to-process or process-to-process .It is generally in terms of attributes. There may be either an input data flow or output data flow.

### Functional processing:



The process is nothing but the transformation of data . It starts with the subject and has the verb followed by the subject.

### Data store:



It includes file, data base and repository. To parallel lines represent it one end closed rectangle.

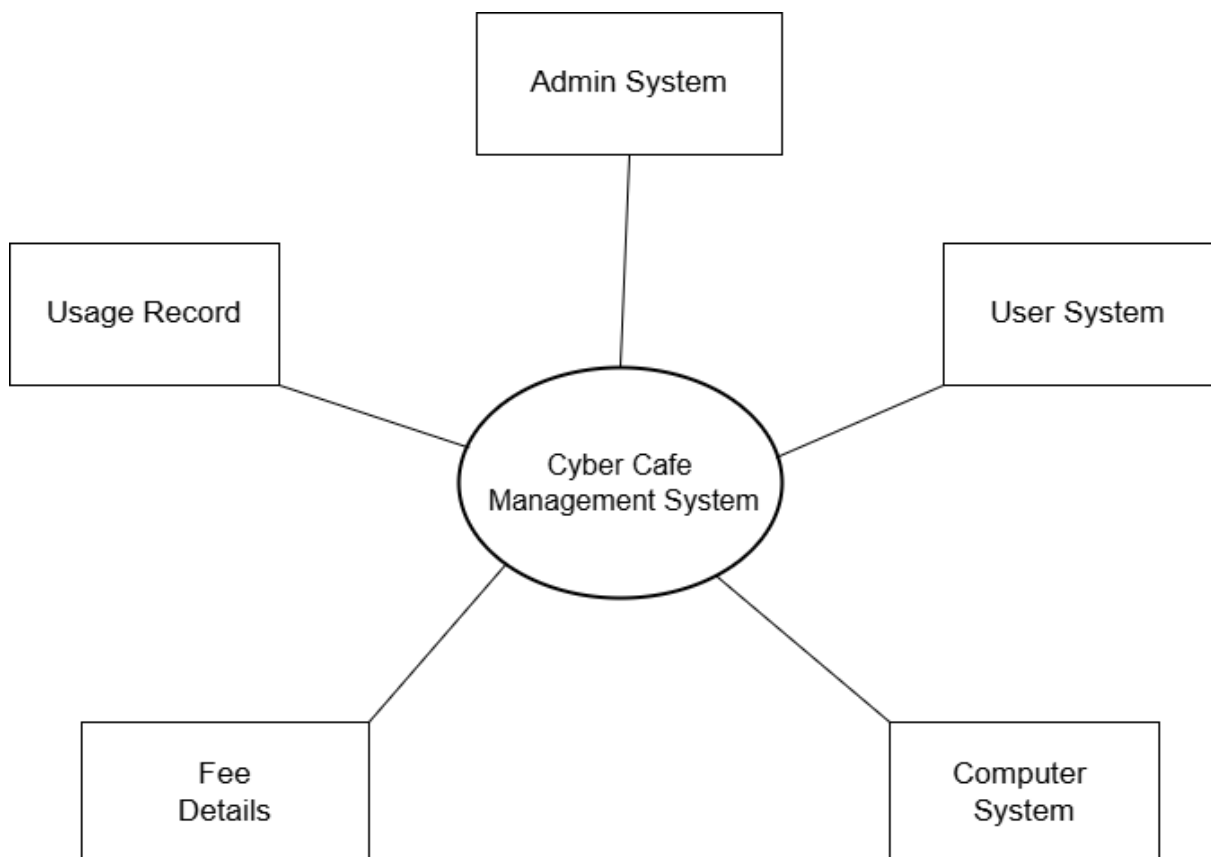
### Actor/source/sink:

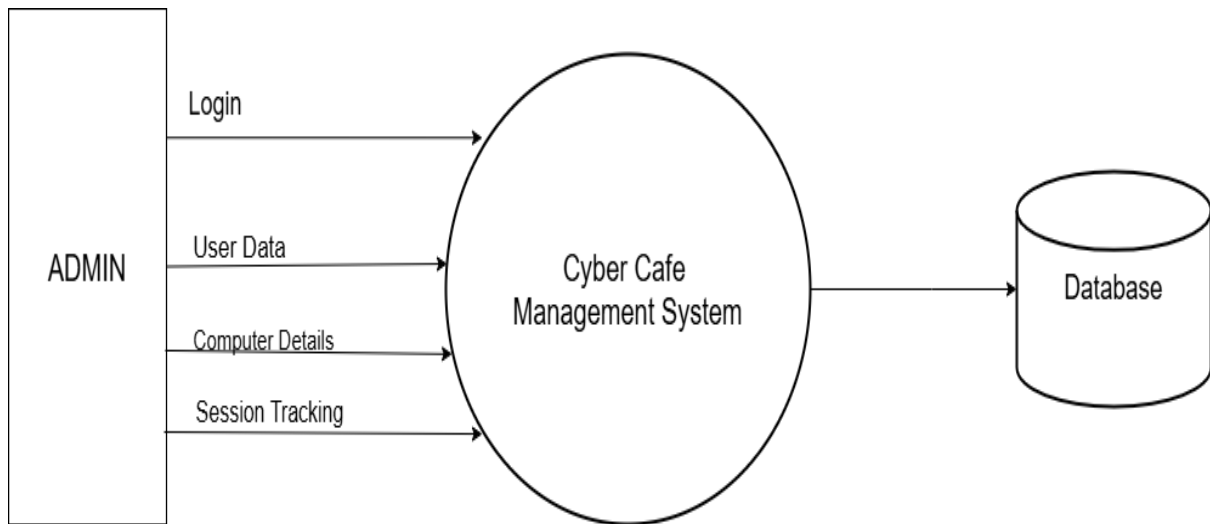


The files which are outside the system and used by the process or processes of the system. Generally Source/sink in the actor.

**Objectives:**

- To graphically document boundaries of a system.
- To provide hierarchy breakdown of the system.
- To show movement of information between a system and its environment.
- To document information flows within the system.
- To aid communication between users and developers.

**Context Model:**

**ZERO LEVEL DATA FLOW DIAGRAM :****ENTITY - RELATIONSHIP MODEL:**

The entity-relationship model is based on a perception of the world as consisting of a collection of basic objects (entities) and relationships among these objects.

- An entity is a distinguishable object that exists.
- Each entity has associated with it a set of attributes describing it.
- E.g. number and balance for an account entity.
- A relationship is an association among several entities.
- e.g. A customer account relationship associates a customer with each account he or she has.

- The set of all entities or relationships of the same type is called the entity set or relationship set.
- Another essential element of the ER diagram is the mapping cardinalities, which express the number of entities to which another entity can be associated via a relationship set.

### **The three main components of ER Diagram are:**

The entity is a person, object, place or event for which data is collected. For example, if we consider the information system for a business, entities would include not only customers, but the customer's address, and orders as well. The entity is represented by a rectangle and labelled with a singular noun.

The relationship is the interaction between the entities. In the example above, the customer places an order, so the word "places" defines the relationship between that instance of a customer and the order or orders that they place. A relationship may be represented by a diamond shape, or more simply, by the line connecting the entities. In either case, verbs are used to label the relationships.

The cardinality defines the relationship between the entities in terms of numbers. An entity may be optional: for example, a sales representative could have no customers or could have one or many customers; or mandatory: for example, there must be at least one product listed in an order. There are several different types of cardinality notations; crow's foot notation, used here, is a common one. In crow's foot notation, a single bar indicates one, a double bar indicates one and only one (for example, a single instance of a product can only be stored in one warehouse), a circle indicates zero, and a crow's foot indicates many. The three main cardinal relationships are: one-to-one, expressed as 1:1; one-to-many, expressed as 1: M; and many-to-many, expressed as M: N.

## Entity Relationship Diagram Notations:

Peter Chen developed ER Diagram in 1976. Since then, Charles Bachman and James Martin have added some slight refinements to the basic ERD principle.

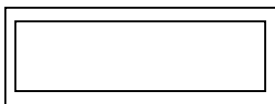
### Entity:

An entity is an object or concept about which we want to store Information. An entity is a real world item or concept that exists on its own. In an ER model, we diagram an entity type as a rectangle containing the type name.



### Weak Entity:

Alone a weak entity is an entity that must be defined by a foreign key relationship with another entity as it cannot be uniquely identified by its own attributes.



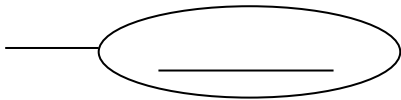
### Attribute:

Each entity has attributes or particular properties that describe the entity. Most of the data in a database consists of values of attributes. The set of all possible values of an attribute is the attribute domain. In an ER model, an attribute name appears in an oval that has a line to the corresponding entity box.



**Key attributes:**

A key attribute is the unique, distinguishing characteristic of the entity. An attribute or set of attributes that uniquely identifies a particular entity is a key. A key attribute in an ER Diagram is represented by an oval that has a line inside it and a line to the corresponding entity box. For example, an employee's social security number might be the employee's key attribute.

**Multi-valued attribute:**

A multi-valued attribute can have more than one value. We indicate this with a double oval. For example, an employee entity can have multiple skill values.

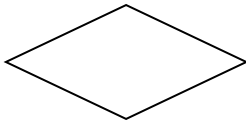
**Derived attribute:**

A derived attribute is based on another attribute. It is denoted by a oval and dotted line within it. For example, an employee's monthly salary is based on the employee's annual salary.

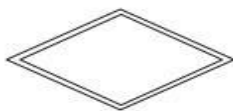


**Relationships:**

Relationships illustrate how two entities share information in the database structure. An association among entities is called a relationship. An attribute can also be a property of a relationship set. The association among the entities is described as one-to-one, one-to-many, many-to-many. A relationship is indicated by a rhombus.

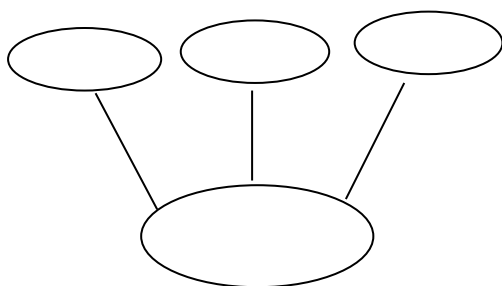
**Identifying relationship:**

Identifying relationship is denoted by double rhombus.

**Composite attribute:**

A composite attribute has multiple components and each component is atomic or composite.

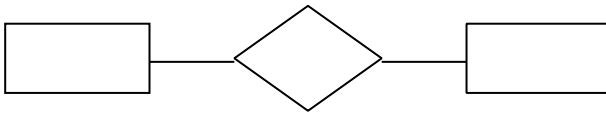
We illustrate this composite nature in the ER model branching of the component attributes.



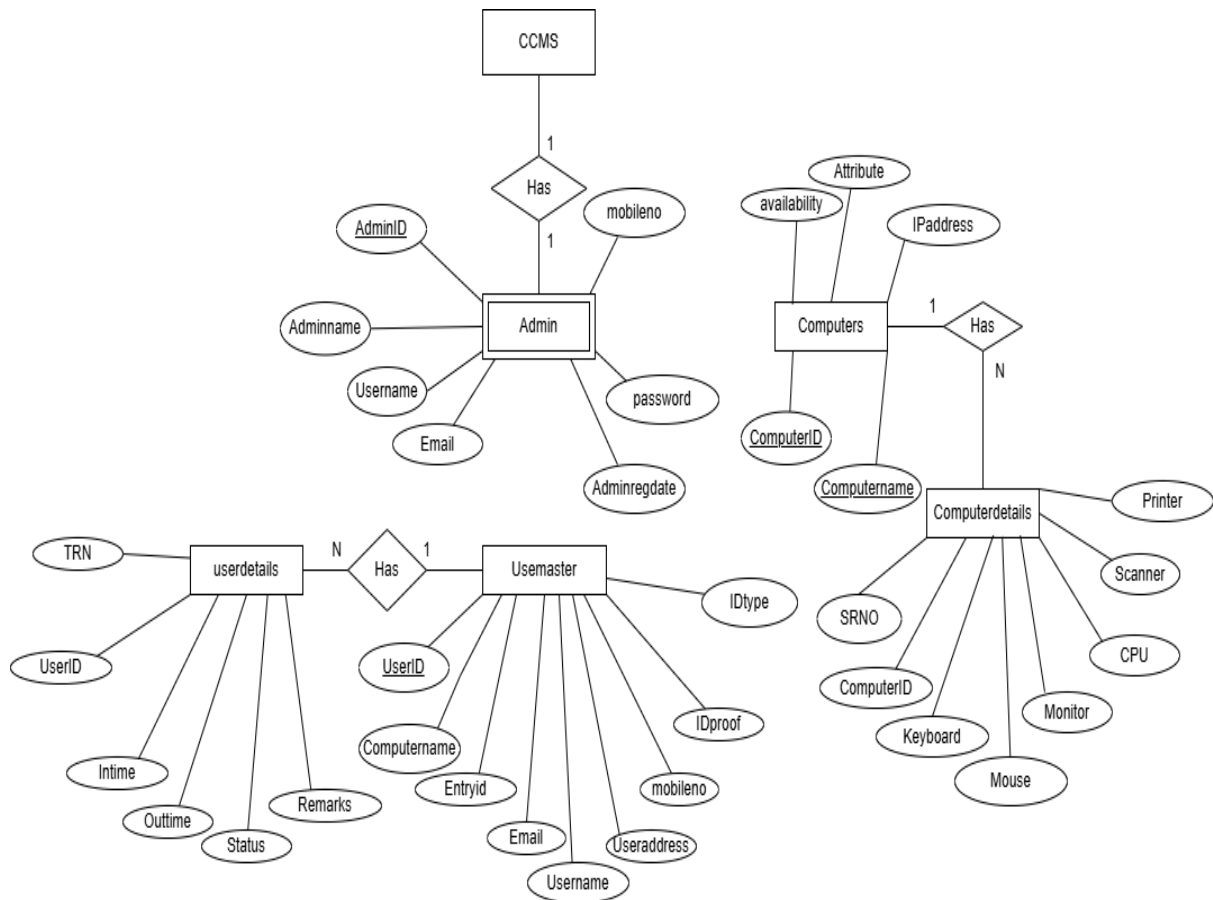
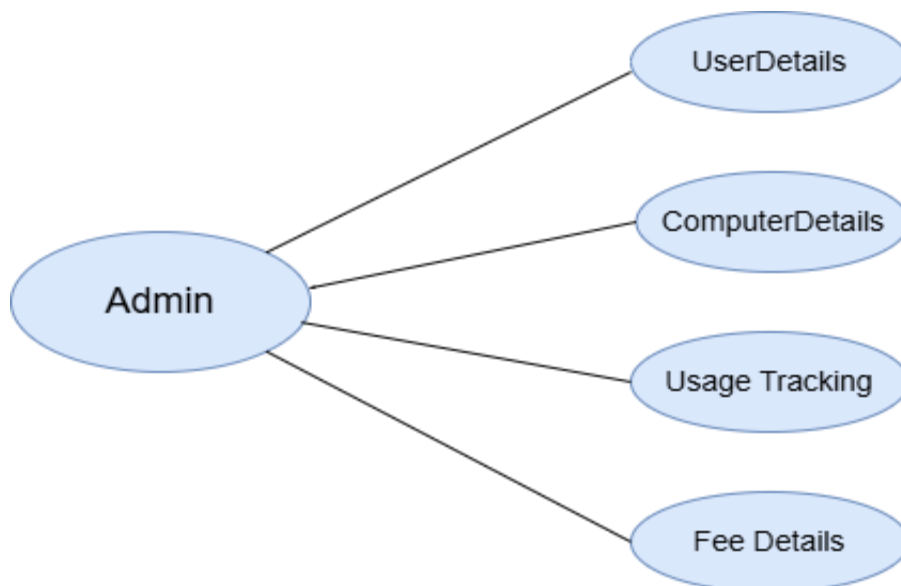


**Total Participation:**

Total participation is represented by a double line.

**Cardinality:**

Cardinality specifies how many instances of an entity relate to one instance of another entity. Ordinarily is also closely linked to cardinality. While cardinality specifies the occurrences of a relationship, ordinarily describes the relationship as either mandatory or optional. In other words, cardinality specifies the maximum number of relationships and ordinarily specifies the absolute minimum number of relationships.

**ER Diagram:****Use Case Diagram:**

## **5.SYSTEM DEVELOPMENT**

### **System Development Phase:**

System development is the process of converting the system design into a functional software application. It involves coding, testing, and integrating various modules to ensure the system meets the specified requirements. The **Cyber Cafe Management System** was developed using a structured, modular approach for better clarity, maintenance, and scalability.

The development phase focuses on how, that is, during development a software engineer attempts to define how data are to be structured, how function is to be implemented within a software architecture, how procedural details are to be implemented, how interfaces are to be characterized and how testing will be performed.

#### **1. Development Approach**

The development followed the **Waterfall Model**, where each phase (Requirement Analysis, Design, Implementation, Testing, and Deployment) was completed in sequence. This model was ideal due to the well-defined and fixed set of requirements for the system.

#### **2. Tools and Technologies Used**

- **Frontend:** HTML, CSS, JavaScript (for forms and user interaction)
- **Backend:** PHP (to handle business logic and database operations)
- **Database:** MySQL (to store member records, transactions, product details)
- **Server:** WAMP/XAMPP (used for local development and testing)
- **Platform:** Windows OS (compatible with WAMP stack)

### **3. Development Steps**

#### **1. Database Creation**

- Tables for admin, users, computer, were created using MySQL.
- Relationships were defined using primary and foreign keys.

#### **2. Frontend Design**

- Web pages were designed with HTML and styled using CSS.
- Forms were created for login, user entry, computer entry, and fee entry.

#### **3. Backend Development**

- PHP scripts were written to handle logic such as admin login, adding users, usage tracking, and fee details.
- Data validation was included to ensure input correctness.

#### **4. Module Integration**

- Each functional module (e.g.. User management, usage tracking) was tested.

### **4. Testing and Debugging**

- Functional testing ensured each feature worked as intended.
  - Sample data was entered to verify usage updates.
  - Bugs and errors were identified and corrected.

## 5. Outcome

The final system is a web-based application that allows the admin to:

- Manage user data
- Record and track usage
- Handle usage session and enter fee details.
- Manage computer details

It offers a simple interface and operates smoothly on standard local server environments.

## **6.SYSTEM IMPLEMENTATION**

System implementation is the process of developing, testing, and deploying the software based on design specifications. For the Cyber Cafe Management System, it involves setting up the environment, creating modules for user, computer management, integrating the database, and ensuring the system runs smoothly for real-time use.

### **1. Objective of Implementation**

To deploy the system in a real-time environment where administrators can perform operations such as managing users, computers and handling usage tracking effectively.

### **2. Implementation Activities**

#### **a. Setting Up the Environment:**

- Installed **WAMP** (Apache, MySQL, PHP) on the development system.
- Deployed the system on a **local server** (WAMP/XAMPP) supporting PHP and MySQL.
- Imported the **MySQL database** using **phpMyAdmin**.
- Configured database connection credentials in PHP files securely.

#### **b. Database Integration:**

- Created necessary tables: admin, computer, user.
- Linked tables using **primary and foreign keys** to maintain data integrity.
- Inserted **sample records** for initial testing of users, computers, usage.

**c. Frontend Deployment:**

- Built the interface using **HTML**, **CSS**, and **JavaScript**.
- Tested the UI for proper layout and responsiveness on desktop browsers.
- Connected forms (login, computer details, user details entry) to backend logic via PHP.

**d. Backend Configuration:**

- Exported the database from the development environment and imported it into the server.
- Set up **PHP scripts** to manage core functions like login, computer details, user details, usage.
- Applied **form validations**, input sanitization, and basic **security measures**.

**3. User Role Setup****Admin:**

- Can log in securely.
- Manage member records (add, update, delete).
- Record usage session.
- Generate fee details.
- Manage Computer details (add, update, remove).

#### 4. User Training

Admin users were trained on:

- How to add and manage users profiles.
- Performing generating fee accurately.
- Viewing and generating reports.
- Managing computer details and updating new one.
- A **user manual with screenshots** was created for easy reference.

#### 5. Post-Implementation Testing

- Tested **login/logout**, user search, and reporting.
- Ensured account balances updated correctly after each transaction.
- Verified product updates reflected accurately in the product catalogue.
- Fixed minor layout and logic issues found during live usage.
- Included a **basic feedback form** to collect admin suggestions.

#### 6. Security Measures Implemented

- Passwords were stored using **secure hashing** methods.
- Admin access was protected through **session handling**.
- Unauthorized access was restricted through session control mechanisms.



## 7. Testing in Live Environment

After deployment, the following were tested:

- Admin login/logout and session control
- Adding and updating users details
- Usage fee generation.
- Data storage and retrieval accuracy

All identified issues were resolved before the final system handover.

## 8. Maintenance and Support

- Regular backups scheduled for the **MySQL database** and application files.
- System logs monitored for unusual activity or errors.
- Support provided through email or local access in case of technical issues.

The implementation of the **Cyber Cafe Management System** was successful. It automated manual operations, improved member, and transaction handling, and provided a more organized and efficient system for cybercafé management. The system is now fully functional and ready for future enhancements based on real-time usage feedback.

## Database tables

### Admin (table)

| # | Name                                                                                             | Type         | Collation          | Attributes | Null | Default           | Comments | Extra             | Action                                                                                                                                                                                                                                                                       |
|---|--------------------------------------------------------------------------------------------------|--------------|--------------------|------------|------|-------------------|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>AdminID</b>  | int          |                    |            | No   | None              |          | AUTO_INCREMENT    |  Change  Drop  More |
| 2 | <b>AdminName</b>                                                                                 | varchar(100) | utf8mb4_0900_ai_ci |            | No   | None              |          |                   |  Change  Drop  More |
| 3 | <b>UserName</b>                                                                                  | varchar(120) | utf8mb4_0900_ai_ci |            | No   | None              |          |                   |  Change  Drop  More |
| 4 | <b>Email</b>                                                                                     | varchar(200) | utf8mb4_0900_ai_ci |            | No   | None              |          |                   |  Change  Drop  More |
| 5 | <b>Passwd</b>                                                                                    | varchar(120) | utf8mb4_0900_ai_ci |            | No   | None              |          |                   |  Change  Drop  More |
| 6 | <b>MobileNumber</b>                                                                              | bigint       |                    |            | No   | None              |          |                   |  Change  Drop  More |
| 7 | <b>AdminRegDate</b>                                                                              | timestamp    |                    |            | Yes  | CURRENT_TIMESTAMP |          | DEFAULT_GENERATED |  Change  Drop  More |

### Computers

| # | Name                                                                                                  | Type         | Collation          | Attributes | Null | Default | Comments | Extra          | Action                                                                                                                                                                                                                                                                             |
|---|-------------------------------------------------------------------------------------------------------|--------------|--------------------|------------|------|---------|----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>ComputerID</b>  | int          |                    |            | No   | None    |          | AUTO_INCREMENT |  Change  Drop  More |
| 2 | <b>ComputerName</b>                                                                                   | varchar(100) | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 3 | <b>ComputerLocation</b>                                                                               | varchar(255) | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 4 | <b>IPAddress</b>                                                                                      | varchar(45)  | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 5 | <b>IsAvailable</b>                                                                                    | tinyint(1)   |                    |            | No   | None    |          |                |  Change  Drop  More |

## Computer Details

| # | Name                                                                                         | Type       | Collation | Attributes | Null | Default | Comments | Extra          | Action                                                                                                                                                                                                                                                                             |
|---|----------------------------------------------------------------------------------------------|------------|-----------|------------|------|---------|----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | SRNO        | int        |           |            | No   | None    |          | AUTO_INCREMENT |  Change  Drop  More       |
| 2 | ComputerID  | int        |           |            | No   | None    |          |                |  Change  Drop  More       |
| 3 | Keyboard                                                                                     | tinyint(1) |           |            | No   | None    |          |                |  Change  Drop  More       |
| 4 | Mouse                                                                                        | tinyint(1) |           |            | No   | None    |          |                |  Change  Drop  More       |
| 5 | Monitor                                                                                      | tinyint(1) |           |            | No   | None    |          |                |  Change  Drop  More       |
| 6 | CPU                                                                                          | tinyint(1) |           |            | No   | None    |          |                |  Change  Drop  More       |
| 7 | Scanner                                                                                      | tinyint(1) |           |            | No   | None    |          |                |  Change  Drop  More       |
| 8 | Printer                                                                                      | tinyint(1) |           |            | No   | None    |          |                |  Change  Drop  More |

## User master

| # | Name                                                                                         | Type         | Collation          | Attributes | Null | Default | Comments | Extra          | Action                                                                                                                                                                                                                                                                             |
|---|----------------------------------------------------------------------------------------------|--------------|--------------------|------------|------|---------|----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Users_id  | int          |                    |            | No   | None    |          | AUTO_INCREMENT |  Change  Drop  More |
| 2 | ComputerName                                                                                 | varchar(100) | utf8mb4_0900_ai_ci |            | No   | None    |          |                |  Change  Drop  More |
| 3 | EntryID                                                                                      | varchar(100) | utf8mb4_0900_ai_ci |            | No   | None    |          |                |  Change  Drop  More |
| 4 | UserName                                                                                     | varchar(100) | utf8mb4_0900_ai_ci |            | No   | None    |          |                |  Change  Drop  More |
| 5 | UserAddress                                                                                  | varchar(200) | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 6 | Email                                                                                        | varchar(200) | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 7 | MobileNumber                                                                                 | varchar(13)  | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 8 | IDProofType                                                                                  | varchar(10)  | utf8mb4_0900_ai_ci |            | Yes  | NULL    |          |                |  Change  Drop  More |
| 9 | IDProofNo                                                                                    | varchar(15)  | utf8mb4_0900_ai_ci |            | No   | None    |          |                |  Change  Drop  More |

**Userdetails**

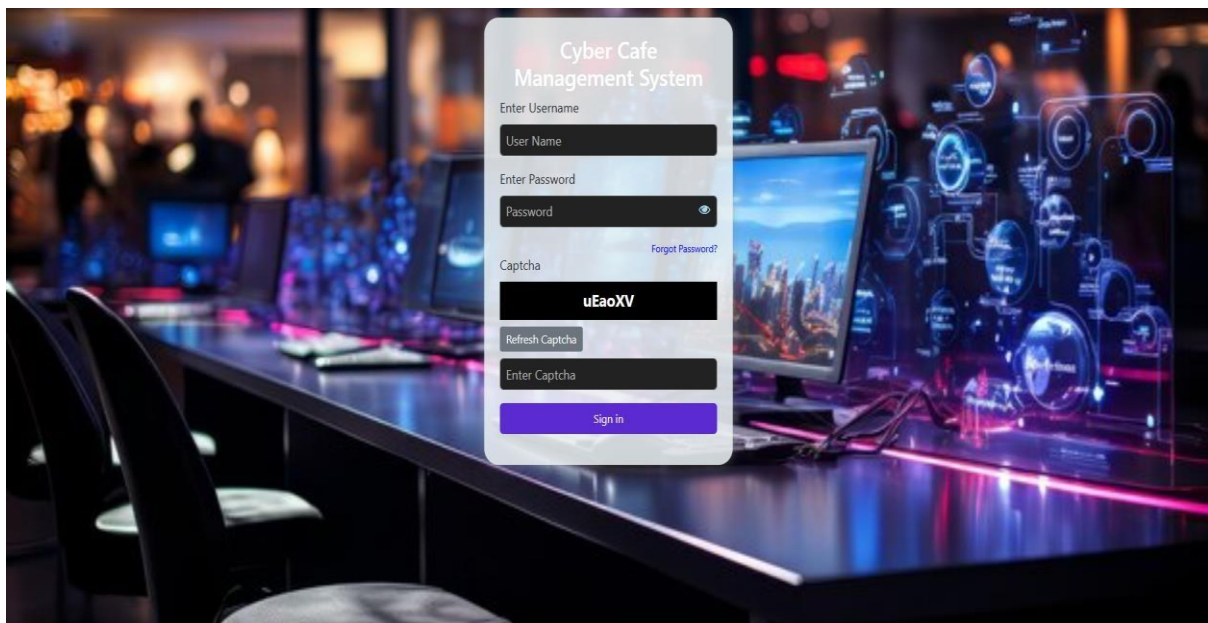
| # | Name                                                                                       | Type        | Collation          | Attributes | Null | Default | Comments | Extra          | Action                                                                                                                                                                                   |
|---|--------------------------------------------------------------------------------------------|-------------|--------------------|------------|------|---------|----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | TRN       | int         |                    |            | No   | None    |          | AUTO_INCREMENT |  Change  Drop More |
| 2 | Users_id  | int         |                    |            | No   | None    |          |                |  Change  Drop More |
| 3 | InTime                                                                                     | timestamp   |                    |            | No   | None    |          |                |  Change  Drop More |
| 4 | TimeOUT                                                                                    | timestamp   |                    |            | No   | None    |          |                |  Change  Drop More |
| 5 | Amount                                                                                     | int         |                    |            | No   | None    |          |                |  Change  Drop More |
| 6 | Remarks                                                                                    | varchar(25) | utf8mb4_0900_ai_ci |            | No   | None    |          |                |  Change  Drop More |
| 7 | status                                                                                     | varchar(20) | utf8mb4_0900_ai_ci |            | No   | None    |          |                |  Change  Drop More |

## 7.SCREEN SHOTS

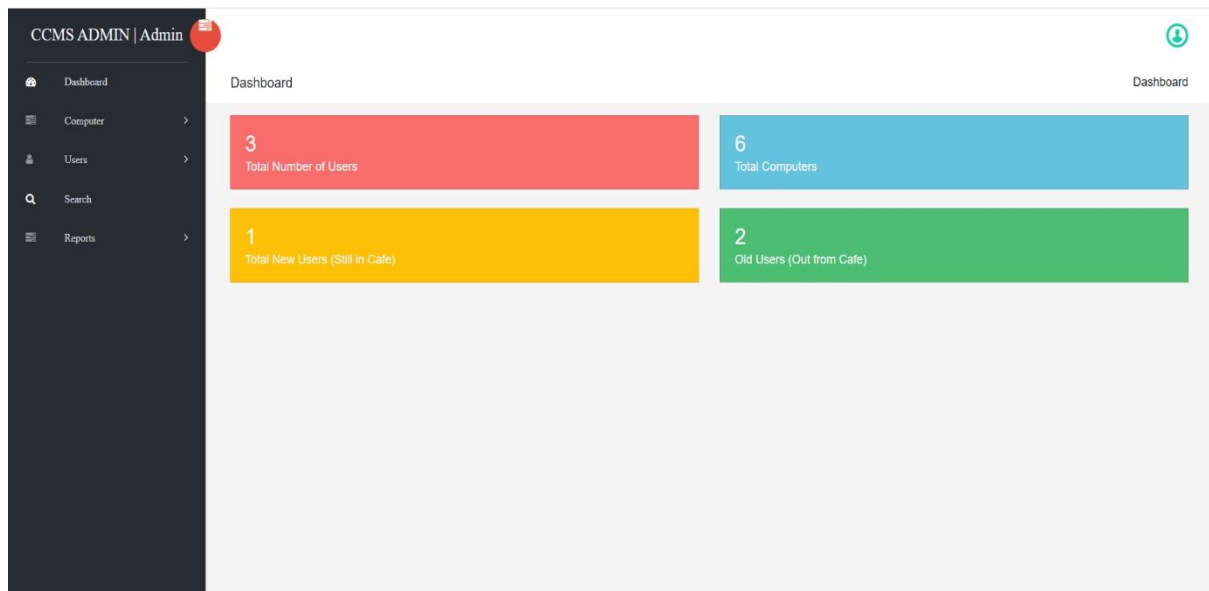
### A view of home page



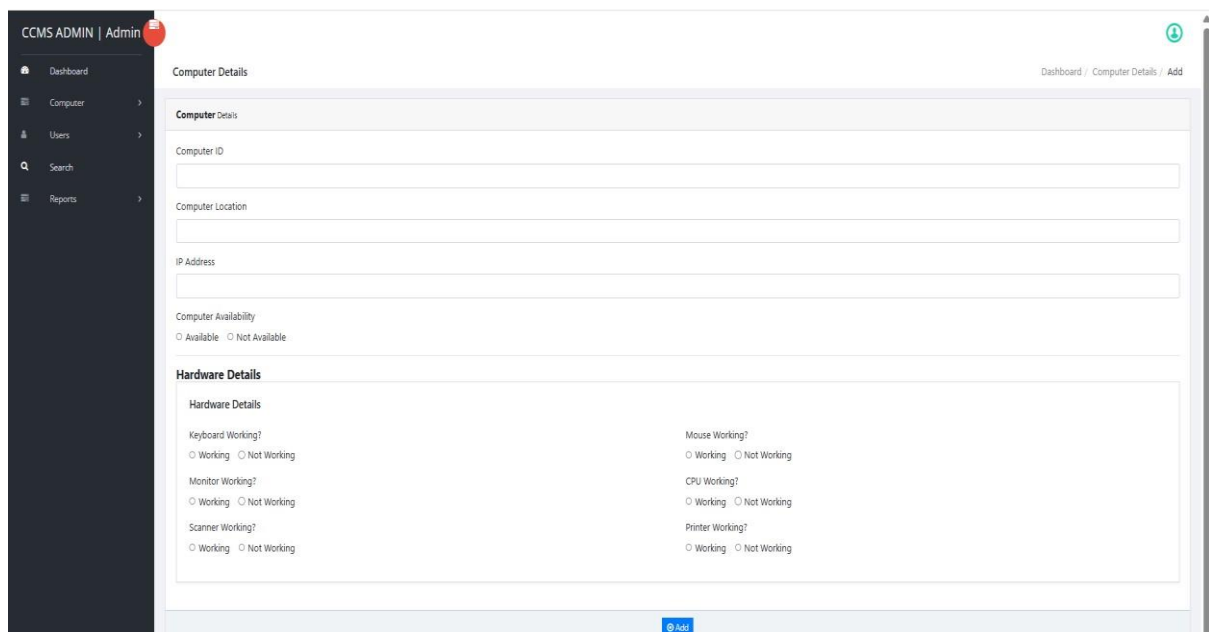
### View of login page



## View of dashboard



## Add computers



The screenshot shows the 'Computer Details' form in the CCMS ADMIN system. The form is divided into two main sections: 'Computer Details' and 'Hardware Details'.

**Computer Details**

- Computer ID:
- Computer Location:
- IP Address:
- Computer Availability: ☐ Available ☐ Not Available

**Hardware Details**

|                                                                                      |                                                                                     |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Keyboard Working?<br><input type="radio"/> Working <input type="radio"/> Not Working | Mouse Working?<br><input type="radio"/> Working <input type="radio"/> Not Working   |
| Monitor Working?<br><input type="radio"/> Working <input type="radio"/> Not Working  | CPU Working?<br><input type="radio"/> Working <input type="radio"/> Not Working     |
| Scanner Working?<br><input type="radio"/> Working <input type="radio"/> Not Working  | Printer Working?<br><input type="radio"/> Working <input type="radio"/> Not Working |

At the bottom right of the form is a blue 'Add' button.

## View of computer details

CCMS ADMIN | Admin

Dashboard / Manage Computers / Manage Computers

### Manage Computers

| S.NO | Computer Name      | Computer Location | IP Address | IsAvailable | Action                                      |
|------|--------------------|-------------------|------------|-------------|---------------------------------------------|
| 1    | Asus               | cabin1            | 7428.4     | 1           | <a href="#">Edit</a> <a href="#">Delete</a> |
| 2    | HP                 | cabin102          | 127.0.0.2  | 1           | <a href="#">Edit</a> <a href="#">Delete</a> |
| 3    | asus               | cabin104          | 234.09     | 1           | <a href="#">Edit</a> <a href="#">Delete</a> |
| 4    | asusbook           | cabin2            | 735.12     | 1           | <a href="#">Edit</a> <a href="#">Delete</a> |
| 5    | DELL               | cabin105          | 127.0.3    | 1           | <a href="#">Edit</a> <a href="#">Delete</a> |
| 6    | Asus gaming laptop | cabin106          | 320.9.7    | 1           | <a href="#">Edit</a> <a href="#">Delete</a> |

## Add user

CCMS ADMIN | Admin

Dashboard / User Detail / Add

### User Detail

**User Details**

User Name

User Address

Mobile Number

Email

Computer Name

ID Proof Type

ID Proof Number

[Add](#)



## View of user details

CCMS ADMIN | Admin

Dashboard / All Users

All Users

Users Details

| S.NO | EntryID   | Full Name     | In Time             | Action               |
|------|-----------|---------------|---------------------|----------------------|
| 1    | 698118448 | Bhavya hegde  | 2025-05-12 12:32:31 | <a href="#">View</a> |
| 2    | 509079373 | varsha        | 2025-05-12 12:42:28 | <a href="#">View</a> |
| 3    | 575822475 | Anirudh singh | 2025-05-12 15:38:30 | <a href="#">View</a> |

## View of fee updating

CCMS ADMIN | Admin

Dashboard / Detailed Info

Detailed Info

|                        |                              |                        |                     |
|------------------------|------------------------------|------------------------|---------------------|
| Entry ID               | 698118448                    | Full Name              | Bhavya hegde        |
| User Address           | kalgaode sirsi uttarakannada |                        |                     |
| Mobile Number          | 8765432567                   | Email                  | bhavya@gmail.com    |
| Computer Name          | HP                           | Computer Location      | cabin102            |
| IP Address             | 127.0.0.2                    | ID Proof               | 32142134 5678       |
| In Time                | 2025-05-12 12:32:31          | Status                 | Check Out           |
| Official/Admin Remarks | <p>Remark:</p> <div>ok</div> |                        |                     |
| Fees                   | 20                           |                        |                     |
| Status:                | Check Out                    |                        |                     |
|                        |                              | <a href="#">Update</a> |                     |
| Remarks                | ok                           | Out Time               | 2025-05-12 18:08:43 |
| Fees                   | 20                           |                        |                     |



## **8.SYSTEM TESTING AND RESULTS**

### **TESTING:**

#### **Introduction**

Testing is a process of executing a program with the intent of finding an error. Testing is a crucial element of software quality assurance and presents ultimate review of specification, design and coding. System Testing is an important phase. Testing represents an interesting anomaly for the software. Thus, a series of testing are performed for the proposed system before the system is ready for user acceptance testing. The code is tested at various levels in software testing. Unit, system and user acceptance testing are often performed.

#### **Testing Objectives**

- Testing is a process of executing a program with the intent of finding an error.
- A good test case is one that has a probability of finding an as yet undiscovered error.
- A successful test is one that uncovers an undiscovered error.

#### **Testing Principles**

- All tests should be traceable to end user requirements.
- Tests should be planned long before testing begins.
- Testing should begin on a small scale and progress towards testing in large.

- Exhaustive testing is not possible.
- To be most effective testing should be conducted by a independent third party.

The primary objective for test case design is to derive a set of tests that has the highest likelihood for uncovering defects in software. To accomplish this objective two different categories of test case design techniques are used. They are:

- White box testing.
- Black box testing.

**White Box Testing:** White box testing focus on the program control structure. Test cases are derived to ensure that all statements in the program have been executed at least once during testing and that all logical conditions have been executed.

**Black Box Testing:** Black box testing is designed to validate functional requirements without regard to the internal workings of a program. Black box testing mainly focuses on the information domain of the software, deriving test cases by partitioning input and output in a manner that provides through test coverage. Incorrect and missing functions, interface errors, errors in data structures, error in functional logic are the errors falling in this category.

**Testing strategies:** A strategy for software testing must accommodate low-level tests that are necessary to verify that all small source code segment has been correctly implemented as well as high-level tests that validate major system functions against customer requirements.

There are two general strategies for testing software. They are as follows:

**Code Testing:** This examines the logic of the program. To follow this test, cases are developed such that every path of program is tested.

**Specification Testing:** Specification Testing examines the specification, starting what the program should do and how it should perform under various conditions. Then test cases are developed for each condition and combinations of conditions and to be submitted for processing.

## Levels of Testing

The stages of Testing Process are:

**Unit Testing:** Individual components are tested to ensure that they operate correctly. Each component tested independently without other system components. Ex. Check for Username and Password with the table, after the next module is loaded.

**Integration Testing:** Integration testing is a systematic technique for constructing the program structure while at the same time conducting test to uncover errors associated with interfacing.

This testing is done using the bottom-up approach to integrate the software components of the software system in to functioning whole.

**System Testing:** System testing is actually a series of different tests whose primary purpose is fully to exercise the computer-based system. The system tests that were applied are recovery testing and performance testing. Finally, a review or audit is conducted which is a final evaluation that occurs only after operating the system long enough for user to have gained a familiarity with it. System testing was done by the inspection team to verify that all the functionality identified in the software requirement specification has been implemented. Defects that crept in the system have been found defect free and is working well. System testing is concerned with interfaces, design logic, control flow recovery, procedures throughput, capacity and timing characteristics of the entire system. For blank field, alphabets, number and special character validation.

**Acceptance Testing:** User acceptance of the system is the key factor for the success of any system. This is done by user. The system is given to the user and they test it with live data. Acceptance testing involves the planning and execution of functional test. Performance tests, stress tests in order to demonstrate that the implemented system satisfies its requirements. Two sets of acceptance test can be run, those developed by the customer. The system has been tested for its performance at unit level by the individuals through performance testing that is designed to test the run time performance of the software. The performance of the fully integrated system is tested and was found good.

### **Test Objectives:**

- The system is tested with variety of inputs. The System is tested for accuracy and correctness of the results obtained. Finally the system is tested for inter-operability.
- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

## Login Form:

| TEST NO | TESTCASE                                                      | EXPECTED RESULT                                        | PASS |
|---------|---------------------------------------------------------------|--------------------------------------------------------|------|
| 1.      | Leave the Username and Password textbox blank and press Login | Prompts to fill the required details                   | Pass |
| 2.      | Enter the Username and Password                               | Homepage will display.                                 | Pass |
| 3.      | Enter invalid Username and Password                           | Error message will display and not direct to next page | Pass |
| 4.      | Leave the text boxes blank in insert and edit form            | Prompt to fill the required details                    | Pass |

### Features to be Tested:

- Verify that the entries are of the correct format.
- No duplicate entries should be allowed.
- All links should take the user to the correct page.

### Future Enhancement:

- Product tracking plan should be done.
- It can be made convenient to interact even through the hand-held devices like tablet and smart phone enabled web site.
- SMS alerts for end users

### Sign up

| TEST NO | TESTCASE                                        | EXPECTED RESULT                      | PASS   |
|---------|-------------------------------------------------|--------------------------------------|--------|
| 1.      | Leave the Form unfilled and press submit button | Prompts to fill the required details | Passed |
| 2.      | Leave one of the fields blank                   | Prompts to fill the required details | Passed |
| 3.      | Fill the required details                       | Inserted record successfully         | Passed |

## **9.LIMITATIONS OF THE PROJECT:**

- **No online payment:** User cannot make online payments through UPI, card or wallet only charges are calculated.
- **No service module:** Extra service like printing, scanning, gaming or lamination are not tracked or charged.
- **No Role based access control:** No extra staff members.

### **Conclusion:**

The Cyber Cafe Management System developed as part of this project successfully automates and streamlines the core operations of a cyber cafe, including user management, computer tracking, and time-based billing. The system ensures efficient handling of users and real-time computer usage monitoring, offering accuracy and reducing manual effort. While the current version includes essential features like user login, session tracking, and charge calculation, it also leaves room for future enhancements such as online payment integration, service modules, invoice generation, and role-based access control.

Overall, this project not only demonstrates practical implementation of database management and PHP-based web development but also serves as a solid foundation for scaling into a more comprehensive commercial solution in the future.

## **10.FUTURE SCOPE AND ENHANCEMENT:**

The Cyber Cafe Management System is designed to be scalable, allowing new features to be added as needed. In the future, the following enhancements can be implemented:

- 1.Online Payment Gateway Integration.
- 2.Service module implementation.
- 3.Mobile version uses for easier access.
- 4.Role based access control



## 11.SOURCE CODE

### **Index page:**

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Cyber Cafe Management System</title>
```

```
  <style>    body {      font-family: Arial, sans-serif;      text-align: center;      background: url('images/welcome.jpg') no-repeat center center/cover;      height: 100vh;      display: flex;      justify-content: center;      align-items: center;      margin: 0;    }
```

```
    .container {      background: rgba(255, 255, 255, 0.8);      padding: 30px;      border-radius: 10px;      box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.2);    } .buttons {      margin-top: 20px;    }
```

```
    .btn {      display: inline-block;      padding: 10px 20px;      margin: 10px;      font-size: 16px;      color: white;      background: #007BFF;
```

```

border: none;      border-
radius: 5px;      cursor:
pointer;          text-
decoration: none;
    }

    .btn:hover      {
background: #0056b3;
    }    p{
color:white;      font-
size:18px;
position:absolute;
bottom:5px;
left:152px;
transform:translate(-
50%);    width:70%;
        text-align:center;
color:rgba(255,255,255,0.5);    text-shadow:2px
2px 4px rgba(0,0,0,0.5);
    }

    .about-btn{      position:
absolute;          top:10px;
right:0%;          transform:
translateX(-50%);    padding:
10px 20px;          font-size: 16px;
color: white;        background:
#a7b6c7;            border: none;
border-radius: 5px;    cursor:
pointer;            text-decoration:
none;
    }

```

```
.about-btn:hover {  
  background: #99b1ca;  
}  
  
.about-section {  display: none; /* Hidden  
initially */  position: absolute;      top: 50%;  
left: 50%;      transform: translate(-50%, -50%);  
background: white;      padding: 30px;  
max-width: 800px;      border-radius: 10px;  
box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.2);  
text-align: center;  
}  
  
.about-section h1 {  
color: #007BFF;  
}  
  
.about-section ul {  
text-align: left;      display:  
inline-block;  
}  
  
.close-btn {  
display: block;  margin-  
top: 20px;  padding: 10px  
20px;  background:  
rgb(99, 64, 197);  color:  
white;  text-decoration:  
none;      border-radius:  
5px;      cursor: pointer;  
}
```

```

</style>                                <script>
document.addEventListener("DOMContentLoaded", function() {
const aboutBtn = document.getElementById("about-btn");    const
aboutSection = document.getElementById("about-section");  const
closeBtn = document.getElementById("close-btn");

    // Clicking "About Us" button shows the section
aboutBtn.addEventListener("click", function(event) {
event.stopPropagation();          aboutSection.style.display = "block";
// Show About Us section
    });

    // Clicking "Close" hides the section
closeBtn.addEventListener("click", function(event) {
event.stopPropagation();          aboutSection.style.display = "none";
// Hide About Us section
    });

    // Clicking anywhere else redirects to login.php
document.body.addEventListener("click", function(event) {    if
(aboutSection.style.display !== "block") {
window.location.href = "login.php";
    }

    });

    });

</script>

</head>

<body>

```

```

<a id="about-btn" class="about-btn">About Us</a>

<div class=""container">

<p>click anywhere to go to the login page</p>

</div>

<div id="about-section" class="about-section">

    <a id="close-btn" class="close-btn">Close</a>

</div>

</body>

</html>

```

## Admin login page:

```

<?php

session_start(); include 'connection.php'; // Database
connection file

// Generate CAPTCHA if not set or refresh is requested if
(!isset($_SESSION['captcha']) || isset($_GET['refresh_captcha'])) {
    $_SESSION['captcha'] =
substr(str_shuffle("ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
123456789"), 0, 6);  if (isset($_GET['refresh_captcha'])) {      echo
$_SESSION['captcha']; // Return new CAPTCHA for AJAX request
    exit;

```

```

    }

}

//      Handle      Login      Submission      if
($_SERVER["REQUEST_METHOD"] == "POST") {
    $username = trim($_POST['username']);

    $password = trim($_POST['password']);

    $captcha = trim($_POST['captcha']);

    // Check CAPTCHA    if ($captcha !=
$_SESSION['captcha']) {        echo "<script>alert('Incorrect
CAPTCHA. Please try again.');"
window.location.href='login.php';</script>";
        exit;
    }

    // Hash the input password using MD5

    $hashed_password = md5($password);

    // Verify user credentials

    $sql = "SELECT AdminID, UserName, Passwd FROM admin WHERE UserName = ?";

    $stmt = $conn->prepare($sql);

    $stmt->bind_param("s", $username);

```

```

$stmt->execute();

$result = $stmt->get_result();

if ($result->num_rows > 0) {

    $row = $result->fetch_assoc();

    // Compare the MD5 hashed password
    if ($hashed_password === $row['Passwd']) {
        $_SESSION['admin_logged_in'] = true;
        $_SESSION['admin_username'] = $username;
        $_SESSION['ccmsaid']=$row['AdminID'];        echo
        "<script>alert('Login Successful!'),
        location.href='dashboard.php';</script>";

        } else {    echo "<script>alert('Invalid Username or
        Password!'),    location.href='login.php';</script>";
        }

        } else {    echo "<script>alert('Invalid
        Username or
        Password!'),location.href='login.php';</script>";
        }

    // Reset CAPTCHA after each login attempt

    $_SESSION['captcha'] =
    substr(str_shuffle("ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0

```

```
123456789"), 0, 6);
```

```
}
```

```
?>
```

```
<!doctype html>
```

```
<html lang="en">
```

```
<head>
```

```
    <title>CCMS Admin Login</title>
```

```
    <link rel="stylesheet" href="vendors/bootstrap/dist/css/bootstrap.min.css">
```

```
    <link rel="stylesheet" href="vendors/font-awesome/css/font-awesome.min.css">
```

```
</head>
```

```
<body>
```

```
    <div class="sufee-login d-flex align-content-center flex-wrap">
```

```
        <div class="container">
```

```
            <div class="login-content">
```

```
                <div class="login-logo">
```

```
                    <div class="heading-container">
```

```
                        <h3 style="color: white">Cyber Cafe Management System</h3>
```

```
                        <hr color="red"/>
```



```
</div>
```

```
</div>
```

```
<div class="login-form">
```

```
<form action="login.php" method="post" autocomplete="off">
```

```
<div class="form-group">
```

```
<label>Enter Username</label>
```

```
<input type="text" class="form-control" placeholder="User Name" required  
name="username" autocomplete="off">
```

```
</div>
```

```
<div class="form-group password-container">
```

```
<label>Enter Password</label>
```

```
<input type="password" class="form-control" placeholder="Password"  
required name="password" id="password" autocomplete="off">
```

```
<i class="fa fa-eye toggle-password" onclick="togglePassword()"></i>
```

```
</div>
```

```
<a href="forgotpassword.php" class="forgot-password">Forgot  
Password?</a>
```

```
<!-- CAPTCHA -->
```

```
<div class="form-group">
```

```
<label>Captcha</label>
```

```

<div id="captcha-text" style="background: black; color: white; font-size:
20px; padding: 8px; text-align: center; font-weight: bold;">

    <?= $_SESSION['captcha']; ?>

</div>

<button type="button" class="btn btn-secondary btn-sm mt-2"
onclick="refreshCaptcha()">Refresh Captcha</button>

<input type="text" class="form-control mt-2" placeholder="Enter Captcha"
required name="captcha">

</div>

<button type="submit" class="btn btn-success btn-flat m-b-30 m-t-30"
name="login">Sign in</button>

</form>

</div>

</div>

</div>

</div>

</div>

<script src="vendors/jquery/dist/jquery.min.js"></script>

<script src="vendors/bootstrap/dist/js/bootstrap.min.js"></script>

```

```

<script>    function
togglePassword() {    var
passwordField =
document.getElementById("pass
word");    var icon =
document.querySelector(".toggle
-password");    if
(passwordField.type ===
"password") {
passwordField.type = "text";
icon.classList.remove("fa-eye");
icon.classList.add("fa-eye-
slash");
    } else {        passwordField.type =
"password";
icon.classList.remove("fa-eye-slash");
icon.classList.add("fa-eye");
    }
}

function refreshCaptcha() {        fetch("captcha.php") // Request new
captcha from the same PHP file
    .then(response => response.text())

    .then(data => {            document.getElementById("captcha-
text").innerHTML = data;
        })

    .catch(error => alert("Failed to refresh CAPTCHA."));

}

```

```
</script>
```

```
</body>
```

```
</html>
```

## Admin profile:

```
<?php
```

```
session_start(); error_reporting(0);
```

```
include('connection.php');
```

```
if (strlen($_SESSION['ccmsaid']) == 0) {
```

```
header('location:logout.php');
```

```
} else { if
```

```
(isset($_POST['submit'])) {
```

```
    $adminid = $_SESSION['ccmsaid'];
```

```
    $AName = $_POST['adminname'];
```

```
    $mobno = $_POST['mobilenumber'];
```

```
    $email = $_POST['email'];
```

```
    $query = mysqli_query($conn, "UPDATE admin SET AdminName='$AName',
```

```
MobileNumber='$mobno', Email='$email' WHERE AdminID='$adminid'");
```

```
    $msg = $query ? "Admin profile has been updated." : "Something Went Wrong. Please try again.";
```

```
    }
```

```
// Fetch the admin data
```

```
$ret = mysqli_query($conn, "SELECT * FROM admin WHERE AdminID='".$_$_SESSION['ccmsaid']."'");
```

```
$row = mysqli_fetch_array($ret);
```

```
// Format the AdminRegDate to YYYY-MM-DD
```

```
$formattedDate = date('Y-m-d H:i:s', strtotime($row['AdminRegDate']));
```

```
}
```

```
?>
```

```
<!doctype html>
```

```
<html class="no-js" lang="en">
```

```
<head>
```

```
<title>CCMS Admin Profile</title>
```

```
<link rel="stylesheet" href="vendors/bootstrap/dist/css/bootstrap.min.css">
```

```
<link rel="stylesheet" href="vendors/font-awesome/css/font-awesome.min.css">
```

```
<link rel="stylesheet" href="vendors/themify-icons/css/themify-icons.css">

<link rel="stylesheet" href="vendors/flag-icon-css/css/flag-icon.min.css">

<link rel="stylesheet" href="vendors/selectFX/css/cs-skin-elastic.css">

<link rel="stylesheet" href="assets/css/style.css">

<link      href='https://fonts.googleapis.com/css?family=Open+Sans:400,600,700,800'
rel='stylesheet'>

</head>

<body>

<?php include_once('sidebar.php'); ?>

<div id="right-panel" class="right-panel">

<?php include_once('header.php'); ?>

<div class="breadcrumbs">

<div class="col-sm-4">

<div class="page-header float-left">

<div class="page-title">

<h1>Admin Profile</h1>

</div>

</div>

</div>

</div>
```

```
<div class="col-sm-8">
```

```
<div class="page-header float-right">
```

```
<div class="page-title">
```

```
<ol class="breadcrumb text-right">
```

```
<li><a href="dashboard.php">Dashboard</a></li>
```

```
<li><a href="adminprofile.php">Admin Profile</a></li>
```

```
<li class="active">Update</li>
```

```
</ol>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="content mt-3">
```

```
<div class="animated fadeIn">
```

```
<div class="row">
```

```
<div class="col-lg-12">
```

```
<div class="card">
```

```
<div class="card-header"><strong>Admin</strong><small>
```

```
Profile</small></div>
```

```

<form name="profile" method="post" action="">

    <p style="font-size:16px; color:red" align="center">

        <?php if ($msg) { echo $msg; } ?>

    </p>

    <div class="card-body card-block">

        <div class="form-group">

            <label class="form-control-label">Admin Name</label>

            <input type="text" name="adminname" value="<?php echo
$row['AdminName']; ?>" class="form-control" required>

        </div>

        <div class="form-group">

            <label class="form-control-label">User Name</label>

            <input type="text" name="username" value="<?php echo
$row['UserName']; ?>" class="form-control" readonly>

        </div>

        <div class="form-group">

            <label class="form-control-label">Contact Number</label>

            <input type="text" name="mobilenumber" value="<?php echo
$row['MobileNumber']; ?>" class="form-control" maxlength="10" required>

        </div>

        <div class="form-group">

```



```

        <label class="form-control-label">Email</label>

        <input type="email" name="email" value="<?php echo
$row['Email']; ?>" class="form-control" required>

    </div>

    <div class="form-group">

        <label class="form-control-label">Admin Registration Date</label>

        <input type="text" value="<?php echo $formattedDate; ?>"
class="form-control" readonly>

    </div>

</div>

<div class="card-footer text-center">

    <button type="submit" class="btn btn-primary btn-sm" name="submit"
id="submit">

        <i class="fa fa-dot-circle-o"></i> Update

    </button>

</div>

</form>

</div>

</div>

</div><!-- .animated -->

</div><!-- .content -->

```

```
</div><!-- /#right-panel -->
```

```
</div>
```

```
<script src="vendors/jquery/dist/jquery.min.js"></script>
```

```
<script src="vendors/popper.js/dist/umd/popper.min.js"></script>
```

```
<script src="vendors/bootstrap/dist/js/bootstrap.min.js"></script>
```

```
<script src="assets/js/main.js"></script>
```

```
</body>
```

```
</html>
```

## **12.BIBLIOGRAPHY**

### **References**

1. *Software Engineering* – 9th Edition by Ian Sommerville
2. *Fundamentals of Database Systems* – B. Navathe
3. Ivan Bayross (2010). *Web Enabled Commercial Application Development Using HTML, JavaScript, DHTML and PHP*. BPB Publications.
4. Achyut S. Godbole & Atul Kahate (2008). *Web Technologies: TCP/IP to Internet Application Architectures*. McGraw-Hill Education.
5. *HTML & CSS: The Complete Reference, Fifth Edition*, Thomas A Powell.-2017
6. *MASTERING HTML, CSS & Java Script Web Publishing*, Laura Lemay

### **Websites**

1. [https://www.w3schools.com/php/php\\_intro.asp](https://www.w3schools.com/php/php_intro.asp) – Web development tutorials and references
2. <https://getbootstrap.com/docs/5.3/getting-started/introduction/> – Bootstrap framework for responsive design
3. <https://www.youtube.com/@CodeWithHarry/playlists> Code with harry Youtube channel
4. <https://chat.openai.com> – ChatGPT AI assistance for coding and explanation
5. <https://deepseek.com> – AI-based development help and coding suggestions

**AI Tools Used:**

During the development and documentation of the LocalHandz project, the following AI tools and platforms were utilized to improve efficiency, assist with code generation, and enhance content writing

1. ChatGPT(OpenAI):

Used extensively for code suggestions, system design drafting, technical documentation writing, testing strategy planning, and troubleshooting PHP-MySQL integration issues. It also assisted in refining report sections like system design, development, testing, and user manuals.

2. GitHub Copilot:

Integrated within the code editor (VS Code), GitHub Copilot provided real-time code completion and intelligent suggestions for HTML, CSS, JavaScript, and PHP. It significantly accelerated backend logic implementation and frontend validations.

3. Canva AI (for diagrams):

Assisted in creating clean and visually appealing architecture and system design diagrams using AI-powered templates and layouts.